



Knowledge Graphs

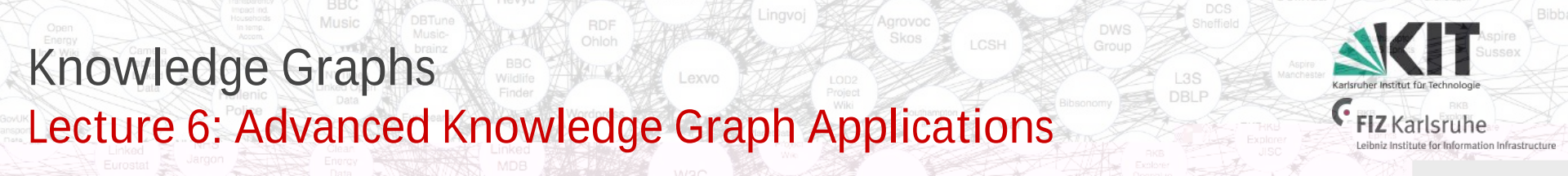
Lecture 6 - Advanced Knowledge Graph Applications

6.6 Exploratory Search and Recommender Systems

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Knowledge Graphs

Lecture 6: Advanced Knowledge Graph Applications

6.6 Exploratory Search and Recommender Systems

Retrieval vs. Exploration

The Retrieval Problem

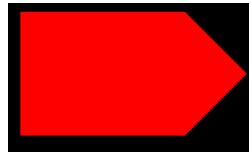
- Retrieval Problem:
 - something specific
- specify your search request
 - *author name, title, etc.*
- -
 -

Author: Jules Verne

Title: From the Earth to the Moon



The Retrieval Problem



V E R N E, Jules:
From the Earth to the Moon, Direct in
97 Hours 20 Minutes and a Trip Round It,
Sampson Low, Marston & Company,
London (1873),
viii, 323 p. plates.

GRC C.194.a.6

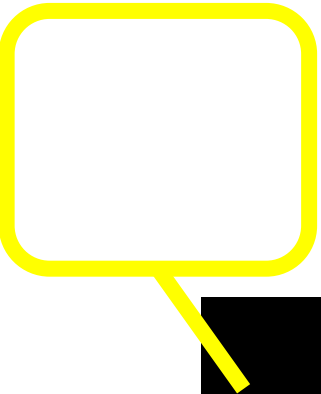
Retrieval vs. Exploration



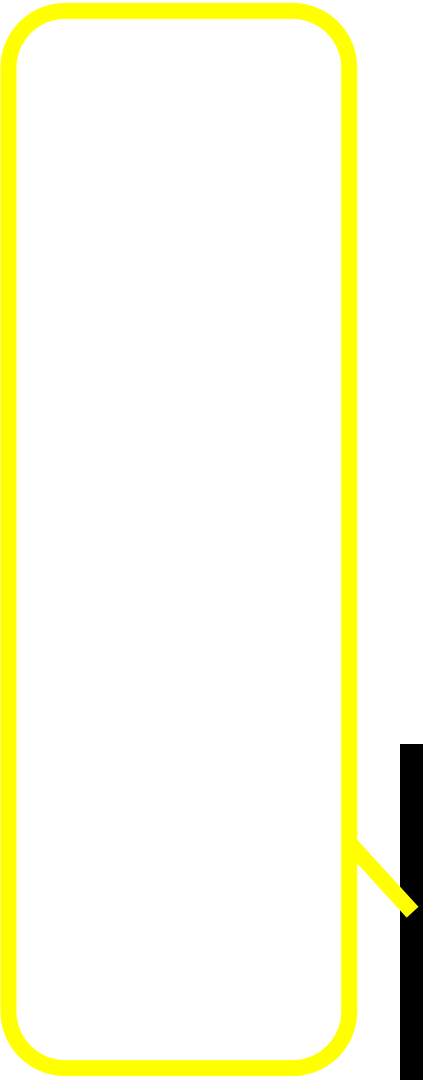
- *Find another („comparable“) book, (that will be of interest for me...)*
- *Find books of the same or of related topics*
- *How did the author / the topic develop over time?*
- *What else would I like to read?*
- ...



Exploratory Search



...and intelligent
recommendations

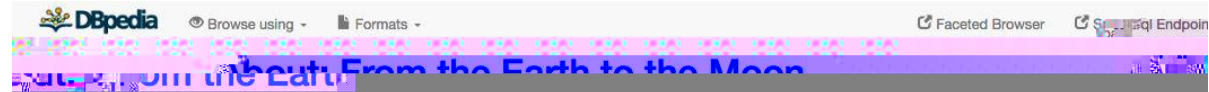


Traditional Libraries
enable Exploratory Search

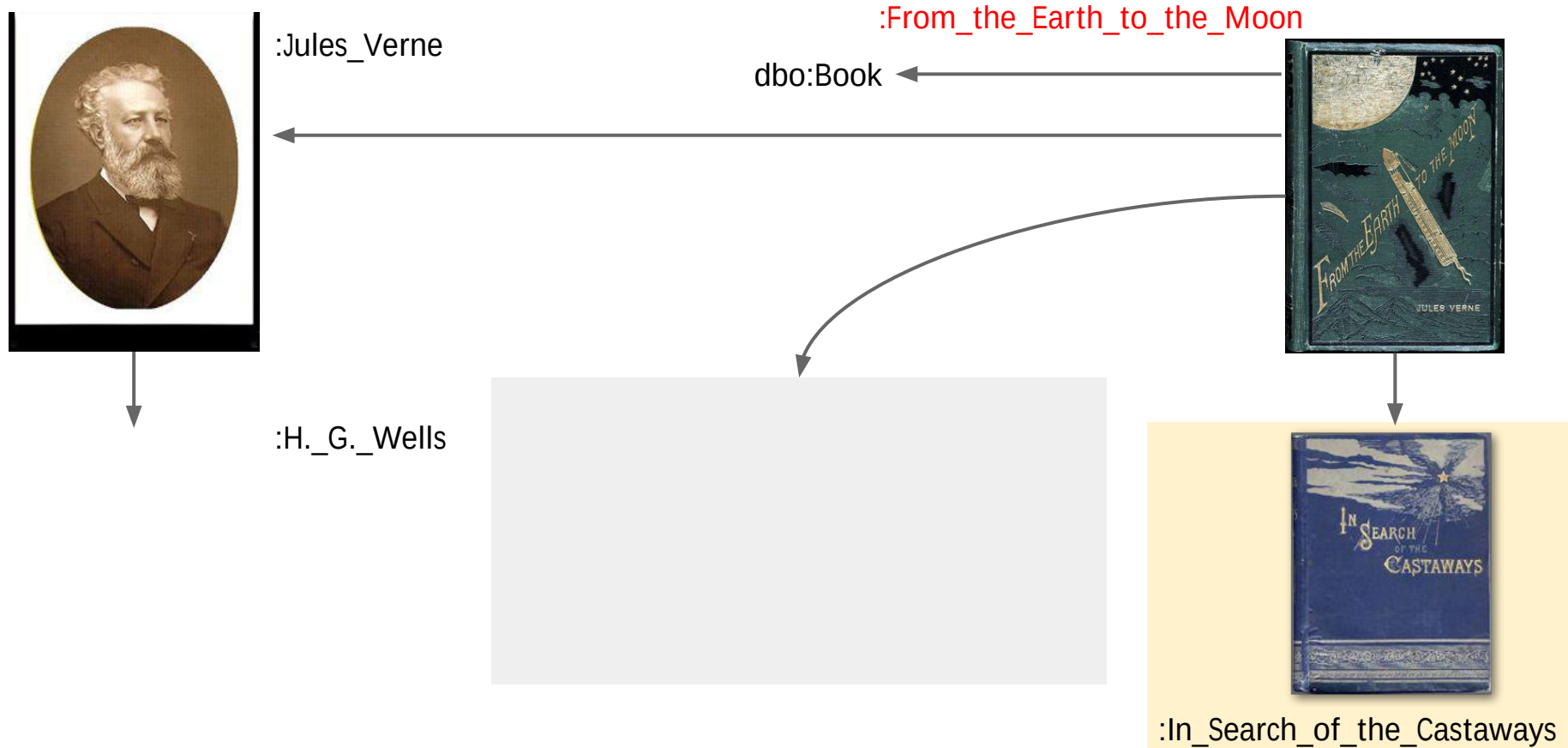
Exploratory Search and Recommendation



Exploratory Search via Knowledge Graphs



Exploratory Search via Knowledge Graphs

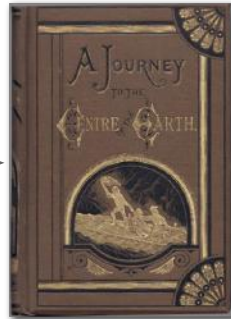


Exploratory Search via Knowledge Graphs

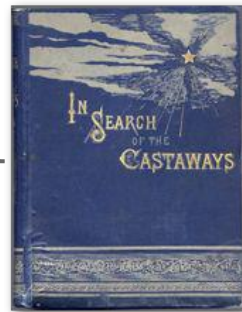
:From_the_Earth_to_the_Moon



dbo:Book



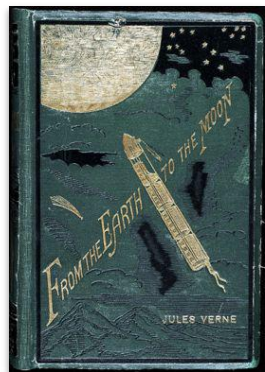
:A_Journey_to_the_Center_of_the_Earth



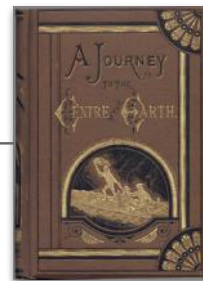
:In_Search_of_the_Castaways

Exploratory Search via Knowledge Graphs

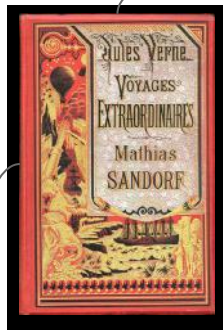
:From_the_Earth_to_the_Moon



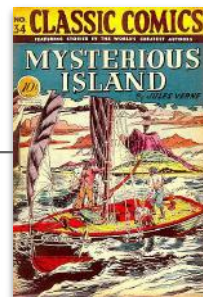
dbo:Book



:A_Journey_to_the_Center_of_the_Earth



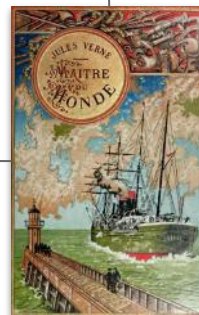
:Matthias_Sandorf



:The_Mysterious_Island



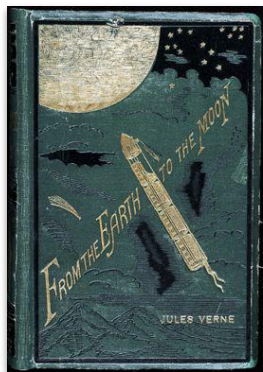
:Jules_Verne



:Master_of_the_World_(novel)

Exploratory Search via Knowledge Graphs

:From_the_Earth_to_the_Moon



dbo:Book

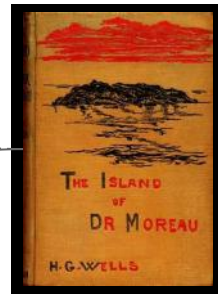
dbo:Writer



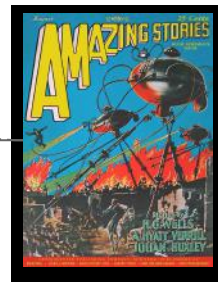
:Jules_Verne



:The_Invisible_Man



:The_Island_of_Doctor_Moreau



:The_War_of_the_Worlds

:H._G._Wells

Exploratory Search via Knowledge Graphs

- Exploratory Search
 - unfamiliar with the domain
 - unsure about the ways
 - unsure about their goals
- Recommender Systems

Knowledge Graph Based Recommendations

Virtuoso SPARQL Query Editor

Default Data Set Name (Graph IRI)

[About](#) | [Namespace Prefixes](#) | [Inference rules](#) | [RDF views](#) | [SPARQL](#)

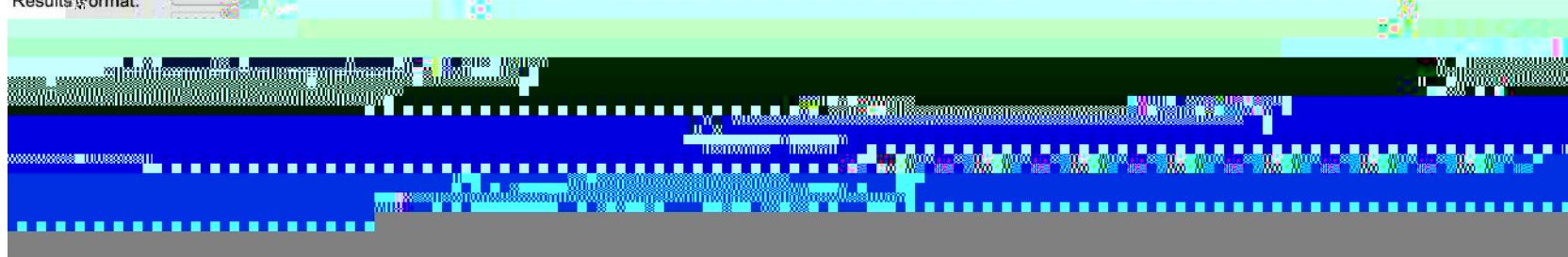
Query Text

```
SELECT ?s (COUNT(DISTINCT(?category)) AS ?score)
WHERE {
  dbr:From_the_Earth_to_the_Moon dct:subject ?category.
  ?s dct:subject ?category FILTER (?s != dbr:From_the_Earth_to_the_Moon) .
} GROUP BY ?s
ORDER BY DESC (?score)
```

(Security restrictions of this server do not allow you to retrieve remote RDF data, see [details](#).)

Results Format:

HTML



...and please ~~don't~~ make better suggestions!



**Thank you very much for
your Attention!**

Picture References:

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